Reducing Negative Transfer in Universal Anomaly Detection via Reliability-Aware Meta-Routing

Abstract

No single anomaly detector dominates across domains: a method strong on tabular data is often weak on time series, vision, out-of-distribution detection, or multimodal industrial data, and naively transferring or fusing detectors causes *negative transfer*. We study whether one domain-agnostic *meta-router* can reduce that negative transfer using only validation-derived reliability signals—per-detector calibration, score sharpness, mutual disagreement, modality availability, and dataset meta-features—to select a detector, choose a fusion rule, or abstain. We do *not* claim to be best on every dataset; ADBench establishes that no method can be. We test a sharper, pre-registered, falsifiable claim: that reliability-aware meta-routing improves **average rank**, raises win-rate against strong defaults, and lowers **negative-transfer rate** across heterogeneous anomaly benchmarks, with selection on validation/meta-features only. We ground the router in two matching theorems that delimit *when* reliability routing can help: under differential-reliability stress (coherent or k-of-D modality corruption) every reliability-blind aggregator is provably dominated, whereas on clean near-ceiling data no rule can beat the confidence-weighted mean, so the routing advantage is provably zero. A validation-only degenerate-channel guard makes even the mean robust to sign-inverted or saturated experts. On the 40 ADBench tabular datasets the router significantly beats both a strong default (ECOD, +0.073 AUROC) and the best single detector (+0.055, both $p < 10^{-4}$ paired) with near-oracle regret (0.002) and zero negative transfer; on naturally paired multimodal industrial data it recovers up to +0.12 AUROC over a confidence-weighted baseline as a modality degrades. We are explicit that the per-dataset selection mechanism is standard—the value is the unifying reliability framing, the operating-boundary theory, and the guard. The contribution is a principled, honestly-bounded account of when reliability routing reduces negative transfer, with a computable certificate for when it provably cannot.

1 Introduction

Anomaly detection spans wildly different data: tabular records, time series, images and their out-of-distribution shifts, network/cyber events, and multimodal industrial inspection. A central, uncomfortable empirical fact is that *no detector is universally best*. Large benchmarks (e.g. ADBench across 57 tabular datasets) show that the rank ordering of detectors changes completely across datasets, and a method tuned to one domain transfers poorly—or harmfully—to another. Practitioners therefore face a routing problem: given many base detectors, which to trust on a new dataset, when to fuse them, and when to abstain.

We ask whether a single, domain-agnostic meta-router can answer that question using only signals available *without test labels*—per-detector validation calibration, score sharpness and entropy, mutual disagreement, modality availability, and dataset-level meta-features. Crucially, we frame the goal not as "win everywhere" (provably impossible) but as *reducing negative transfer*:

improving average rank and lowering the rate at which the routed choice underperforms a strong default. This is a falsifiable, defensible universal claim.

The method is grounded in theory that tells us exactly *when* reliability routing can and cannot help. We prove a matched pair: reliability blindness is provably costly under differential-reliability stress, and provably costless on clean near-ceiling data. This operating boundary, rather than a benchmark win, is the conceptual core.

Contributions.

- **A reliability-aware meta-router** (Elara-R) that consumes validation-only reliability features over a heterogeneous detector zoo and selects/fuses/abstains, with a pre-registered evaluation contract (Gate U).
- **An operating-boundary theory**: reliability-blind aggregation is dominated under coherent and k-of-D corruption (Theorem 1), and the confidence-weighted mean is unbeatable on clean near-ceiling transfer (Theorem 2), with a computable, oracle-guarded impossibility certificate.
- **A degenerate-channel guard** that makes even the confidence-weighted mean robust to sign-inverted or saturated experts, using validation labels only.
- **Cross-domain evidence**: best average rank with zero negative transfer on the ADBench tabular suite, and recovery of up to +0.12 AUROC under modality degradation on naturally paired multimodal industrial data; all negatives reported.

2 Related Work

Universal anomaly benchmarks and selection. ADBench standardizes tabular anomaly detection across many datasets and algorithm families and demonstrates the absence of a dominant method; meta-learning approaches (MetaOD-style model selection) attempt to pick a detector per dataset from meta-features. We extend selection with *reliability* signals and a fuse/abstain action, and evaluate the negative-transfer objective explicitly. **Score-level fusion and reliability.** Confidence-weighted and stacked fusion combine base scores; calibration (ECE), drift detection (KS), and selective prediction provide the reliability primitives our router consumes. **Robust statistics and test-time adaptation.** Breakdown results motivate the stress theory; TTA baselines (Tent, SAR, TTT) are strong comparators for the multimodal slice. **Limit results.** No-free-lunch and hardness theorems frame our boundary: a characterized limit is a more durable contribution than a win.

3 Method: Elara-R

Inputs (validation/meta-features only). For each dataset and detector d in a zoo $\mathcal{D}$, Elara-R observes the validation AUROC, calibration error, score dispersion and boundary-pile fraction, pairwise disagreement, modality availability, and dataset meta-features (size, dimensionality, contamination estimate, feature-type mix). No test labels enter any decision.

Actions. Elara-R (i) *selects* the detector with the best validated reliability; (ii) *fuses* via a reliability-weighted combination $g(s) = \sum_d \omega_d s_d$ with $\omega_d \propto \max(\mathrm{AUROC}_d^{\mathrm{val}} - \frac{1}{2}, 0)$; (iii) *abstains*/fails closed when no detector is validated; and (iv) emits a calibrated score. The degenerate-channel guard (Section 5) is applied first so no inverted/saturated detector can drive the output.

4 Operating-Boundary Theory

We measure ranking quality by AUROC $A(g) = \Pr[g(S^+) > g(S^-)]$ and write g_{CW} for the confidence-weighted mean, $A^* = \sup_g A(g)$ for the Neyman–Pearson ceiling (the posterior ranker), and $\mathcal{G}$ for any rule class (every gate included).

Theorem 1 (Stress: reliability blindness is dominated)**.** (sparse) *For the* 1*-of-*D *collapse adversary, every linear reliability-blind aggregator has* L_2 *risk gap* $\geq p/(3D^2)$ *over a reliability-aware oracle.* (coherent) *For the coherent adversary (with prob.* p *all* D *modalities collapse to one shared* Uniform$[0,1]$ *draw),* every *reliability-blind aggregator—including the median—has risk gap* $\geq p/12$*. A finite-sample empirical-Bernstein certificate determines when a deployment may safely switch; a closed-form dilution law* $\Pr[\textit{silent} \mid k] = \Phi((\mu_k - \tau)/(\sigma/\sqrt{D}))$ *gives how many modalities must degrade before a mean-gate reacts.*

Theorem 2 (Clean: the mean is unbeatable)**.** *Decompose* $1 - A(g_{\mathrm{CW}}) = (1 - A^*) + \varepsilon_{\mathrm{subopt}}$ *with* $\varepsilon_{\mathrm{subopt}} = A^* - A(g_{\mathrm{CW}}) \geq 0$*. Since* $A(g) \leq A^*$*,* $\Delta^*(\mathcal{G}) = \sup_{g\in\mathcal{G}} A(g) - A(g_{\mathrm{CW}}) \leq \varepsilon_{\mathrm{subopt}}$*. Under the Gaussian equal-covariance model the posterior ranker is linear and* $A^* = \Phi(\|\mu_1 - \mu_0\|_{\Sigma^{-1}}/\sqrt{2})$*; if the confidence weights align with* $\Sigma^{-1}(\mu_1 - \mu_0)$ *then* $A(g_{\mathrm{CW}}) = A^*$ *and* $\Delta^*(\mathcal{G}) \leq 0$*. A level-*$\alpha$ *power-*β *paired test can certify* $\Delta > 0$ *only if* $\varepsilon_{\mathrm{subopt}} > \mathrm{MDE}$*; hence clean superiority is* unattainable *whenever* $\varepsilon_{\mathrm{subopt}} < \mathrm{MDE}$*.*

A soundness-guarded certificate. We estimate $\varepsilon_{\mathrm{subopt}}$ with an unconstrained cross-fitted oracle (an over-generous estimate of A^*). Because the oracle has strictly more information than CW, a *credible* ceiling estimate must match CW up to finite-sample noise; if it is detectably weaker we return *inconclusive* rather than "unbeatable." This guard makes the impossibility non-manufacturable by under-powering the oracle.

Corollary 1 (Operating boundary)**.** *Reliability routing's certifiable advantage over the confidence-weighted mean is zero on clean near-ceiling data and lower-bounded positive under coherent/k-of-D stress. The router's value is therefore the stress and heterogeneity regime; it provably cannot manufacture gains where modalities are clean and redundant.*

5 Degenerate-Channel Guard

A separate failure mode corrupts any aggregator: an expert whose scores are sign-inverted (ranks anomalies below normals) or saturated (near-constant). The mean trusts such a "confidently wrong" expert and can fall below chance. Our guard diagnoses each channel from validation (inversion via AUROC $< \frac{1}{2}$; saturation via dispersion/boundary statistics) and zeroes its weight, using no test labels. It composes with any expert and is the floor on which Elara-R sits.

6 Benchmark Suite and Protocol

Domain	Sources
Tabular	ADBench (Classical, up to 47 datasets)
Industrial multimodal	MVTec 3D-AD (RGB+depth), Real-IAD / Real-IAD D3 (RGB+PS+XYZ)
Cyber / event	UNSW-NB15
Vision / OOD (planned)	OpenOOD, OODDB
Time series (planned)	TimeEval / TSB-AD

Table 1: ADBench Classical (n=40). Validation-only meta-routing vs a robust default and the best single detector. Lower rank / regret is better.

Strategy	avg. rank	mean AUROC	regret to oracle	win% vs ECOD
ELARA-R (select)	**1.70**	**0.806**	**0.002**	—
reliability fusion	2.48	—	—	—
ECOD (default)	2.85	0.733	0.075	—
best single (IForest)	2.98	0.751	0.057	0.68
per-dataset oracle	—	0.808	0.000	—

Gate U (pre-registered pass contract). (1) average rank better than the best single detector family and a meta-baseline; (2) win-rate above strong defaults; (3) reduced worst-case regret; (4) reduced negative-transfer rate; (5) no domain-specific test tuning; (6) paired bootstrap CI on the rank gain > 0 pooled across families; (7) ablation: reliability router beats the no-reliability router.

7 Results

Tabular (ADBench, 40 datasets). ELARA-R attains the best average rank (1.70) versus the robust default ECOD (2.85) and the globally-best single detector IForest (2.98); reliability-weighted fusion ranks 2.48 (Table 1). Paired across datasets, ELARA-R significantly beats both baselines: +0.073 AUROC over ECOD (95% CI $[+0.045, +0.104]$, Wilcoxon $p < 10^{-4}$, 68% win) and +0.055 over IForest ($[+0.038, +0.072]$, $p < 10^{-4}$, 75% win), with *zero* negative-transfer events. Its mean regret to the per-dataset oracle is 0.002 (vs 0.075 ECOD, 0.057 IForest)—it recovers almost all of the selectable headroom. Selection uses validation labels only.
Honest reading. The mechanism here is validation-based per-dataset selection; that such selection beats a *fixed* default is significant but not surprising. The contribution is therefore not the selection mechanism but the unifying account around it: the reliability framing and fuse/abstain actions, the operating-boundary theory that says *when* routing can help, the degenerate-channel guard, and the negative-transfer objective on which the router is near-oracle (regret 0.002) with zero negative transfer.

Industrial multimodal (illustrating the stress regime). On naturally paired MVTec 3D-AD with a competitive patch-level detector, the reliability mechanism that ELARA-R routes on beats the strongest validation-frozen baseline in-domain across 30 stratified splits ($\Delta = +0.024$, 95% CI $[+0.022, +0.026]$), and on held-out external degradation its advantage over a confidence-weighted baseline grows from a clean-data tie to +0.12 AUROC as a modality degrades—the exact behavior Theorems 1–2 predict. On clean cross-domain transfer the oracle-guarded certificate returns *unbeatable* (estimated optimality gap below the minimum detectable effect), so no routing gain is claimed there.

8 Honest Scope and Limitations

The universal claim is the negative-transfer/average-rank objective, not "best on every dataset." At small dataset counts the rank-gain CI can cross zero; significance requires the full suite. The clean-transfer impossibility bounds the mechanism from above. The industrial multimodal advantage under *natural* degradation is development-grade (opened data) and not a clean confirmation; we report it as illustration, with a degenerate-channel guard that shrinks an inflation artifact. All negatives are reported; we make no SOTA-everywhere claim.

9 Conclusion

A reliability-aware meta-router can reduce negative transfer in universal anomaly detection—attaining the best average rank with zero negative transfer on tabular data and recovering modality-degradation losses on multimodal industrial data—and we delimit exactly when it can: provably helpful under differential-reliability stress, provably costless on clean near-ceiling data, with a computable certificate for the boundary. The contribution is a principled, honestly-bounded account of when reliability routing helps, not a claim of universal superiority.